

Senario Based Practice Problems [Important for Mid]

Lets learn how to read a question. Most students without reading the whole question directly jump into the code, which is completely wrong. So, how you maximize the efficiency? When you get the question paper in your hand read the whole question in the first place. Lets say, you have 75 Minutes. Allocate 5 minutes for reading the whole question. Mark the key findings (see Q1 for example). Highlight those findings. Once you done with reading the whole question, you already highlighted what the question asked from you. Now think of what tools you need to solve the problem (tools: conditions -> if, else if, else | loops -> for/while, while operators you need). Once you decoded everything, now attempt the easiest problem. Don't try to just pass the given input case, think of your code should work for any given input accordingly to the given question. All the best!

1. An e-commerce site offers discounts during a promotional event. Purchases over 2,000 BDT get a 10% discount. Prime members get an additional 5% discount on the amount after the initial discount is applied, regardless of the purchase amount. A flat 50 BDT delivery fee is added to the final amount if the total before the delivery fee is less than 1,000 BDT. Now, write a program that asks for the user's total purchase amount and whether they are a "Prime" member. Calculate the final bill. Every member has a unique ID, which denotes them as a prime member or not.

2. A ride-sharing app calculates fares based on the vehicle type and distance. Moto: The base fare is 25 BDT, plus 12 BDT per kilometre. Car: The base fare is 50 BDT, plus 22 BDT per kilometre. CNG: The base fare is 30 BDT, plus 15 BDT per kilometre. During "peak hours" (4 PM to 7 PM), a 20% surcharge is applied to the total fare (base fare + distance charge). Ask the user for the distance of the ride, the vehicle type, and whether it is a peak hour. Calculate and print the total fare.

3. A fitness app tracks a user's daily calorie intake. The user has a daily goal of 2,200 calories. Write a program that prompts the user to enter their calorie intake for each of the 7 days of a week. The program should then calculate the total and average weekly calorie

intake. Finally, it should count and display how many days the user was over, under, or exactly at their daily goal.

4. A coffee shop has a digital menu. The user first chooses a category, then an item from that category. Coffee: Latte (220 BDT), Cappuccino (200 BDT), Americano (180 BDT) Tea: Green Tea (100 BDT), Lemon Tea (80 BDT). Snacks: Sandwich (250 BDT), Croissant (150 BDT). Create a menu-driven program where the user first selects a category (1 for Coffee, 2 for Tea, 3 for Snacks). Based on the choice, display the items in that category and ask them to choose an item. Finally, print the selected item and its price.

5. A small shop needs a program to check its inventory. The reorder point for any item is 20 units. The desired stock level is 75 units. The program should repeatedly ask the shopkeeper for an item's current stock count. If the count is below the reorder point of 20, it should print a "Reorder required" message and calculate how many units need to be ordered to reach the desired stock level of 75. If the stock is sufficient, it should print "Stock level is OK." The program should only stop when the user enters -1.

6. A university library uses a 7-digit student ID for its system. A valid ID must meet two criteria: the sum of its digits must be greater than 25, and the ID must not contain the digit '0'. Ask a user to input their 7-digit ID. Write a program to loop through the digits of the number to check if it meets both criteria and then print whether the ID is "Valid" or "Invalid".

7. An event has 4 sessions. The organiser wants to find out which session was the most popular. Write a program that ask for the number of attendees for each of the 4 sessions. After getting the input for all sessions, the program should determine which session had the highest attendance and print the session number and its attendee count.

8. A mobile operator wants to help customers choose the right monthly data plan. Usage less than 3 GB: "Mini Pack (50 BDT). Usage from 3 GB to 10 GB: "Standard Pack (150 BDT)". Usage from 11 GB to 30 GB: "Mega Pack (300 BDT)". Usage more than 30 GB: "Unlimited Pack (500 BDT)". Ask the user for their estimated average monthly data usage in GB and recommend the appropriate plan and its price.

9. A cinema hall has 7 rows and 10 seats in each row. Write a program that first displays the entire seating arrangement using the character 'A' for Available. Then, prompt the user to enter a row and seat number they wish to book. Finally, display the seating arrangement again, but this time showing the user's selected seat as 'B' for Booked.

10. An ATM allows a user to perform multiple actions after successfully entering their PIN. Assume the PIN is correct. Present the user with a menu: 1. Check Balance, 2. Withdraw, 3. Deposit, 4. Exit. Use a do-while loop to allow the user to perform multiple transactions. The loop should continue until the user selects the "Exit" option. (You can use a dummy balance of 50,000 BDT).

11. A bank has a policy for personal loans: the applicant must earn at least 40,000 BDT per month and must not have any other existing loans. Ask the user for their monthly income and whether they have any existing loans (yes/no). Use branching with logical operators to determine and print if they are "Eligible for a loan" or "Not eligible for a loan".

12. A factory produces widgets in batches. Each widget is given a quality score. The program needs to categorise the widgets from a single batch.

- Score > 95: "Excellent"
- Score 85-95: "Good"
- Score 70-84: "Acceptable"
- Score < 70: "Defective"

Now, First, ask the user for the number of widgets in a batch. Then, loop that many times, asking for the quality score of each widget. The program should count how many widgets fall into each category and print a summary report at the end.

13. A courier service calculates shipping costs based on package weight and destination type. Base cost for all packages: 60 BDT. If the weight is over 10 kg, add an extra 100 BDT. If the destination is "local," there's no extra charge. If it's "domestic," add a 50 BDT surcharge. Ask the user for the package weight and destination type ("local" or "domestic"). Calculate and print the total shipping cost based on the combination of rules.

14. A vending machine offers 3 items: Item A (20 BDT), Item B (35 BDT), and Item C (50 BDT). The machine only accepts 10, 20, and 50 BDT notes. Ask the user to first enter the amount of money they are inserting. Then, ask them to select an item (A, B, or C). The program must check if the inserted amount is sufficient. If it is, it should print "Item dispensed" and calculate and display the change. If not, it should print "Insufficient funds."

15. A website requires a new password to meet certain criteria for security.

- Weak: Fails to meet two or more criteria.
- Medium: Meets two of the three criteria.
- Strong: Meets all three criteria.

The criteria are

1. Must be at least 8 characters long.
2. Must contain at least one number (0-9).
3. Must contain at least one uppercase letter (A-Z).

Now, Ask a user to enter a password. Loop through the password to check for the presence of numbers and uppercase letters. Based on the checks, classify the password as "Weak," "Medium," or "Strong."

16. A high-security digital lock has a unique way of generating a one-time passkey. To open it, a user enters a 5-digit primary code, and the system's passkey is the exact reverse of that code.

Prepared and Organized by
BADHON NANDI
Student Tutor & FYAT Mentor
Department of Computer Science and Engineering, Brac University

Editing in progress

Email me for any question additions/enquiries: badhon.nandi@g.bracu.ac.bd